HW7

In slides

7-(1) A one bit error is corrected because the syndrome provides a unique address that points directly to the specific bit position that flipped; once the position is identified, the error is fixed by simply flipping that bit back to its original state.

7-(2) Programmable Array Logic is a type of PLD to implement complex digital circuits by featuring a programmable array followed by a fixed OR array. This specific architecture makes PALs easier to program and more cost-effective than other devices as only the products terms are customizable while the summation remains predefined by hardware.

Problem:

7.12(a) 000011101010

$C_1(1,3,5,7,9,11)=0$ V

$C_2(2,3,6,7,10,11)=1$ x

$C_4(4,5,6,7,12)=1$ x

$C_8(8,9,10,11,12)=0$ V

origin: 01011010

7.19

x \ yz	00	01	11	10
0		1	1	
x 1		1		1

$A = x'z + y'z + xyz'$

x \ yz	00	01	11	10
0	1	1		
x 1			1	1

$B = x'y' + xy$

x \ yz	00	01	11	10
0			1	
x 1		1		

$C = x'yz + xy'z$

x \ yz	00	01	11	10
0		1		1
x 1	1	1	1	

$D = xy' + xz + y'z + x'yz'$

	term	x	y	z	A	B	C	D
$x'z$	1	0	-	1	1	-	-	-
$y'z$	2	-	0	1	1	-	-	1
xyz'	3	1	1	0	1	-	-	-
$x'y'$	4	0	0	-	-	1	-	-
xy	5	1	1	-	-	1	-	-
$x'yz$	6	0	1	1	-	-	1	-
$xy'z$	7	1	0	1	-	-	1	-
xy'	8	1	0	-	-	-	-	1
xz	9	1	-	1	-	-	-	1
$x'yz'$	10	0	1	0	-	-	-	1

7.25

x \ yz	00	01	11	10
0		1		1
x 1	1			1

$A = yz' + xz' + x'y'z$

x \ yz	00	01	11	10
0	1	1	1	
x 1	1		1	

$B = x'y' + y'z' + yz$

x \ yz	00	01	11	10
0		1		1
x 1	1		1	1

$C = xz' + xy + yz' + x'y'z$

$= xyz + A$

x \ yz	00	01	11	10
0		1	1	1
x 1		1	1	

$D = z + x'y$

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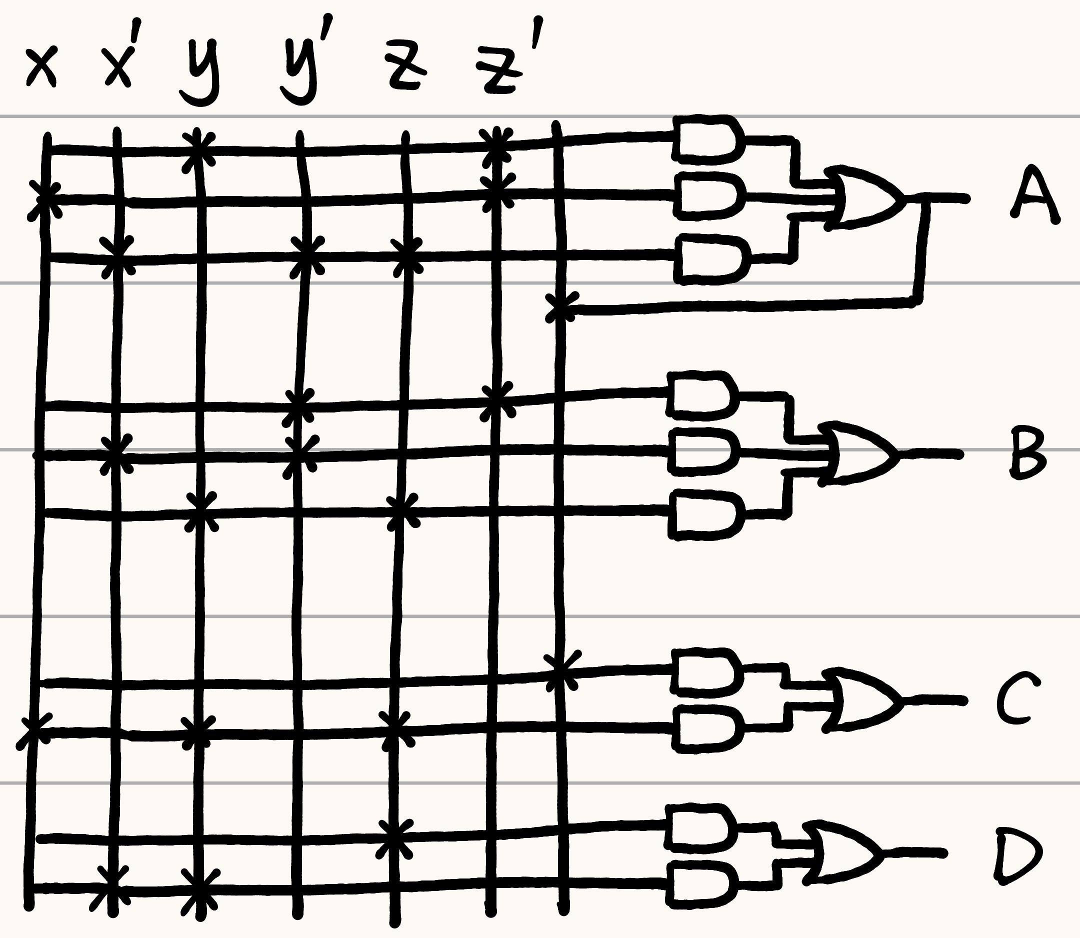
	term	x	y	z	A
yz'	1	-	1	0	-
xz'	2	1	-	0	-
$x'y'z$	3	0	0	1	-
$y'z'$	4	-	0	0	-
$x'y'$	5	0	0	-	-
yz	6	-	1	1	-
A	7	-	-	-	1
xyz	8	1	1	1	-
z	9	-	-	1	-
$x'y$	10	0	1	-	-

$A = yz' + xz' + x'y'z$

$B = y'z' + x'y + yz$

$C = A + xyz$

$D = z + x'y$



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